

Chapter (20)

Database-System Architectures

I. Speedup, Scaleup

1. A program has 80% of its execution time that can be parallelized. If the program is executed using 4 processors, calculate the maximum speedup using Amdahl's Law. Determine whether the speedup is linear or sublinear.
2. A parallel system uses 10 processors. If 90% of the execution time is parallelizable, calculate the scaleup using Gustafson's Law. Determine whether the scaleup is linear or sublinear.

II. Atomic Instruction

test-and-set and compare-and-swap

3. Suppose three processes P1, P2, and P3 access a shared memory location M representing an exclusive lock. Initially, $M = 0$. P1 executes Test-and-Set(M) first, followed by P2 and P3. Determine the value read, the value returned, the value set in M, and whether each process acquires the lock. Explain how Test-and-Set provides mutual exclusion.
4. Suppose three processes P1, P2, and P3 access a shared memory location M representing an exclusive lock. Initially, $M = 0$. Each process executes compare-and-swap (M, 0, 1) to acquire the lock. If P1 executes first, determine the value read, whether the comparison succeeds, the value set in M, and whether each process acquires the lock. Explain how Compare-and-Swap provides mutual exclusion.

Chapter (21)

Parallel and Distributed Storage

I. Partitioning Strategies

1. Consider a relation with the following tuples

Tuple	Employee ID
T1	101
T2	102
T3	103
T4	104
T5	105
T6	106
T7	107
T8	108

The relation is partitioned across **4 nodes: N1, N2, N3, N4**. Determine the node on which each tuple is stored using:

- (a) Round-robin partitioning
- (b) Hash partitioning, where $h(ID) = ID \bmod 4$
- (c) Range partitioning with partition vector [103, 105, 107].

II. Skew in Partitioning

2. A relation contains 1,000 tuples and is partitioned across 10 nodes. Ideally, each node should contain $1000/10=100$ tuples. However, because of data distribution skew, one node contains 200 tuples while the other nodes contain fewer tuples.
 - (a) Calculate the expected speedup.
 - (b) Calculate actual speedup
 - (c) Explain the effect of skew.

Chapter (22)

Parallel and Distributed Query Processing

I. Parallel Processing

1. What form of parallelism is likely to be the most important for each of the following tasks?
 - (a) Three independent queries Q1, Q2, and Q3 are executed simultaneously on different nodes.
 - (b) A single query is divided into several parts, and different parts of the same query are processed simultaneously on multiple nodes.
 - (c) A query contains a sort operation, and the sorting operation itself is divided among several nodes so that different portions of the data are sorted in parallel.
 - (d) A query performs a selection operation and a join operation. The join can start processing tuples while the selection operation is still producing its output.
 - (e) Several independent queries submitted by different users are executed concurrently by the database system.

II. Parallel Join

2. Consider the following relations. Perform the join using the asymmetric fragment-and-replicate join. Show the fragmentation, replication, local join results, and final result.

Employee

EmployeeID	DeptID
1	10
2	20
3	10
4	30

Department

DeptID	DeptName
10	IT
20	HR
30	Finance

III. Map and Reduce Operations

3. Consider the following input:

Database is fast

Database is distributed

Using the MapReduce model, perform the following:

- (a) Apply the map() function and produce the intermediate key-value pairs.
- (b) Group all records with the same intermediate key.
- (c) Apply the reduce () function.
- (d) Give the final output.

Chapter (23)

Parallel and Distributed Transaction Processing

I. Commit Protocol

1. Consider a distributed transaction T using the Two-Phase Commit protocol. The coordinator has failed after all participating nodes have written <ready T> to their logs, but before any node has received <commit T> or <abort T>.
 - (a) What is the state of transaction T?
 - (b) Can the participating nodes decide to commit or abort immediately?
 - (c) What problem occurs in this situation?
 - (d) What must the nodes do?
2. A distributed transaction T is executed at three participating nodes: N1, N2, and N3. During Phase 1 of the Two-Phase Commit (2PC) protocol, N1 and N2 reply READY, but N3 replies ABORT.
 - (a) What decision will the coordinator make?
 - (b) Will transaction T be committed or aborted at all nodes?
 - (c) Explain why.
3. Consider a distributed transaction T using the Two-Phase Commit protocol. The coordinator has failed after all participating nodes have written <ready T> to their logs, but before any node has received <commit T> or <abort T>.
 - (a) What is the state of transaction T?
 - (b) Can the participating nodes decide to commit or abort immediately?
 - (c) What problem occurs in this situation?
 - (d) What must the nodes do?

II. Concurrency Control in Distributed Database

4. A node in a distributed system obtains an exclusive lease from a lock manager at 10:00:00 AM for 30 minutes. At 10:25:00 AM, the node requests a lease renewal, and the lock manager grants an additional 20 minutes.
 - (a) Determine the original lease expiry time.
 - (b) Determine the new lease expiry time after the renewal.
 - (c) Determine the total lease duration.
 - (d) What happens if the node fails before the renewed lease expires?
 - (e) Why is lease expiration useful in a distributed system?

5. Consider the following local wait-for relationships:

- Site S1: $T1 \rightarrow T2$
- Site S2: $T2 \rightarrow T3$
- Site S3: $T3 \rightarrow T4$
- Site S1: $T4 \rightarrow T1$

(a) What does the edge $T1 \rightarrow T2$ mean?

(b) Construct the global wait-for graph.

(c) Determine whether a deadlock in a distributed system. Give a reason.

(d) Identify the cycle in the global wait-for graph.

Chapter (25)

Advanced Application Development

I. Performance Tuning

1. The following hardware features are used in the configuration of a database system:
 - An SSD costs \$150 and supports 480 random I/O operations per second (IOPS)
 - Main memory costs \$100 for 64 GB. The database page size is 16 KB
 - (a) Calculate the break-even time (m) for a page.
 - (b) Should a page accessed once every 3 hours be kept in memory?
2. An application performs 2,000 random reads per second and 500 random writes per second. Each disk supports 100 I/O operations per second. Calculate:
 - (a) The total I/O operations per second required by RAID 5.
 - (b) The total I/O operations per second required by RAID 1.
 - (c) Which configuration requires fewer disks?

II. Performance Bottleneck

3. Two database systems, A and B, handle two types of transactions. System A has throughputs of 99 transactions/sec and 1 transaction/sec, while System B has throughputs of 50 transactions/sec and 50 transactions/sec. Assuming that both transaction types are equally likely:
 - (a) Calculate the average throughput of System A using the harmonic mean.
 - (b) Calculate the average throughput of System B using the harmonic mean.
 - (c) Determine which system is faster and by how many times.

Chapter (8)

Complex Data Types

I. RDF and Knowledge Graphs

1. Consider the following two documents:

Document D1

Parallel databases divide large datasets across multiple processors to execute queries faster. They improve performance by distributing query operations simultaneously. Fault tolerance and load balancing are important to ensure consistent results in parallel systems.

Document D2

Distributed databases store data across multiple networked sites, allowing users to access information from different locations. Query processing in distributed systems requires coordination between sites to maintain data consistency and reduce latency.

Compute the relevance of each document to the query “Parallel Query” using term frequency and inverse document frequency (TF-IDF).

2. Consider the following Resource Description Framework (RDF) of part of University database.

S101	instance-of	Student.
S101	name	“Alice”.
S101	major	“Computer Science”.
S101	enrolled-in	CS202.
CS202	instance-of	Course
CS202	title	“Data Structures”.
CS202	taught-by	PROF505.
P505	instance-of	Professor.

- (a) Using SPARQL, retrieve the names of all students who enrolled in Data Structures.
- (b) Construct the graph representation of RDF.

II. Object Orientation

3. Acar-rental company maintains a database for all vehicles in its current fleet. For all vehicles, it includes the vehicle identification number, license number, manufacturer, model, date of purchase, and color. Special data are included for certain types of vehicles:
 - Trucks: cargo capacity.
 - Sports cars: horsepower, renter age requirement.
 - Vans: number of passengers.
 - Off-road vehicles: ground clearance, drivetrain (four- or two-wheel drive).

Construct an SQL schema definition for this database. Use inheritance where appropriate.